

Energy Nation Modeling Suite (v2025a)
Comprehensive Counterfactual Modeling Framework for Project-Level Analysis

****Maintainer:**** Joshua Samuel · Northeast Renewables LP
****License:**** CC BY-NC-SA 4.0 — Open for research and educational use. Commercial use prohibited without written permission.

1. Overview

The **Energy Nation Modeling Suite (v2025a)** is a modular pipeline for quantitative evaluation of Canada's Major Projects Inventory (MPI). It integrates risk estimation, expected-value and developer economics, and policy-counterfactual overlays into a unified analytical workflow.

The suite enables transparent, reproducible analysis of how policy signals—such as permitting caps, Indigenous participation, or government override mechanisms—affect project-level probabilities, expected values (EV), and portfolio-level economic outcomes.

2. Dependency Chain

Stage	Script	Description	Key Output
1	`mpi_risk_engine_v03_expanded.py`	Computes base probabilities (Bayes, Cox, blended) for all MPI projects.	`mpi_2024_scored.csv`
2	`ev_dev_engine_5yr.py`	Expands probabilities into 5-year expected-value horizons and developer metrics (PV, MOIC, k*, IRR).	`mpi_2024_ev_dev_combined_5y.csv`
3	`unified_overlay_runner_colab_revised_v2_patched.py`	Applies Indigenous and MPO policy overlays individually and jointly.	Overlay CSVs, summaries, Markdown reports
4	`government_override_runner_colab_v2.py`	Models higher-order government intervention (Override + Contracted-Demand) with α/β grids and interaction θ .	Full per-project and grid summaries

3. Intended Use & Audience

****Intended for:****

- Energy policy analysts, system modelers, and research institutions (e.g., EMH).
- Academic and non-commercial R&D; exploring the effect of policy levers on project viability.

****Not intended for:****

- Commercial forecasting, investment solicitation, or production-grade financial models.

4. Inputs & Outputs

Input	Description	Source
`mpi_2024_input.xlsx`	Base dataset of project features and metadata	NRCan / MPI
`bayes_lr_regenerated_coefficients_expanded.csv`	Log-ratio coefficients for Bayes engine	Derived
`cox_refit_coefficients_timesplit_expanded.csv`	Re-fit Cox regression coefficients	Derived

Output	Description
----- -----	
`mpi_2024_scored.csv`	Base probabilities & urgency metrics
`mpi_2024_ev_dev_combined_5y.csv`	Expected value + developer metrics
`overlay_outputs/*.csv`	Indigenous / MPO overlay results
`overlay_out/*.csv`	Government Override + Contracted Demand results
`.json`, `.md` reports	Audit trail and scenario summaries

5. Parameter Defaults

Category	Parameter	Default
----- ----- -----		
Risk Engine	`BLEND_BAYES_W`	0.50
	`PROVSEC_ALPHA`	0.6
EV / Developer Engine	`EV_HORIZON_YEARS`	5
	`DEV_COST_PCT`	0.03
	`DEV_DISCOUNT`	0.13
	`EV_DISCOUNT`	0.00
	`BASE_YEAR`	2024
MPO / Indigenous Overlays	`APPROVAL_CAP_YEARS`	2
	`TIME_COMPRESSION_YEARS`	-1
	`HAZARD_MULTIPLIER`	1.10
Override / Contracts	`α` grid	0.25–1.00
	`β` grid	0.0–1.0
	`θ` (interaction)	0.00
	`θ_scale`	0.05

6. Methods Summary

- **Risk Engine****:
 - Computes ``p_bayes`` via log-odds product of categorical likelihood ratios with province-sector shrinkage.
 - Computes ``p_cox`` via parametric survival model with re-fit coefficients.
 - Blends the two via fixed weight to produce ``blended_prob``.
- **EV + Developer Engine****:
 - Allocates probability mass across candidate FID years using a softmax allocator.
 - Derives `EV_cum`, `EV_disc`, and developer metrics (``k``, ``PV_MOIC_k``, ``IRR_k``).
- **Unified Overlay Runner****:
 - Applies Indigenous and MPO policy levers as counterfactuals via odds scaling.
 - Computes EV deltas, crossovers, and summary reports.
- **Government Override Runner****:
 - Adds higher-order intervention levers (α , β , θ).
 - Evaluates portfolio uplift, sensitivity bands, and interaction dynamics.

7. Reproducibility

- **Language:** Python 3.10+
- **Libraries:** pandas, numpy, math, argparse, json
- **Execution:** Deterministic for identical CSV inputs.
- **Artifacts:** Each stage writes CSVs with audit-ready metadata and optional JSON/Markdown logs.
- **Suggested structure:**

```
...  
/data/  
/scripts/  
/outputs/  
/docs/  
...
```

8. Limitations & Risks

- Counterfactual overlays are **scenario simulations**, not forecasts.
- EV and probability estimates rely on historical MPI data; extrapolation may not capture post-policy behavior.
- Developer economics assume simplified S-curve spend profiles and constant discount rates.

9. Versioning & Metadata

- **Version:** v2025a
- **Change log:** Adds 5-year horizon, discounted EV basis, and unified overlay structure.
- **Maintainer:** Joshua Samuel · Northeast Renewables LP
- **Contact:** Internal collaboration and R&D; inquiries only.

10. License

Licensed under **Creative Commons Attribution–NonCommercial–ShareAlike 4.0 (CC BY-NC-SA 4.0)**
Open for research and educational use. Commercial use prohibited without written permission.

MODULE CARDS

(Full text for Risk Engine, EV Engine, Overlay Runner, and Government Override Runner as previously shown.)